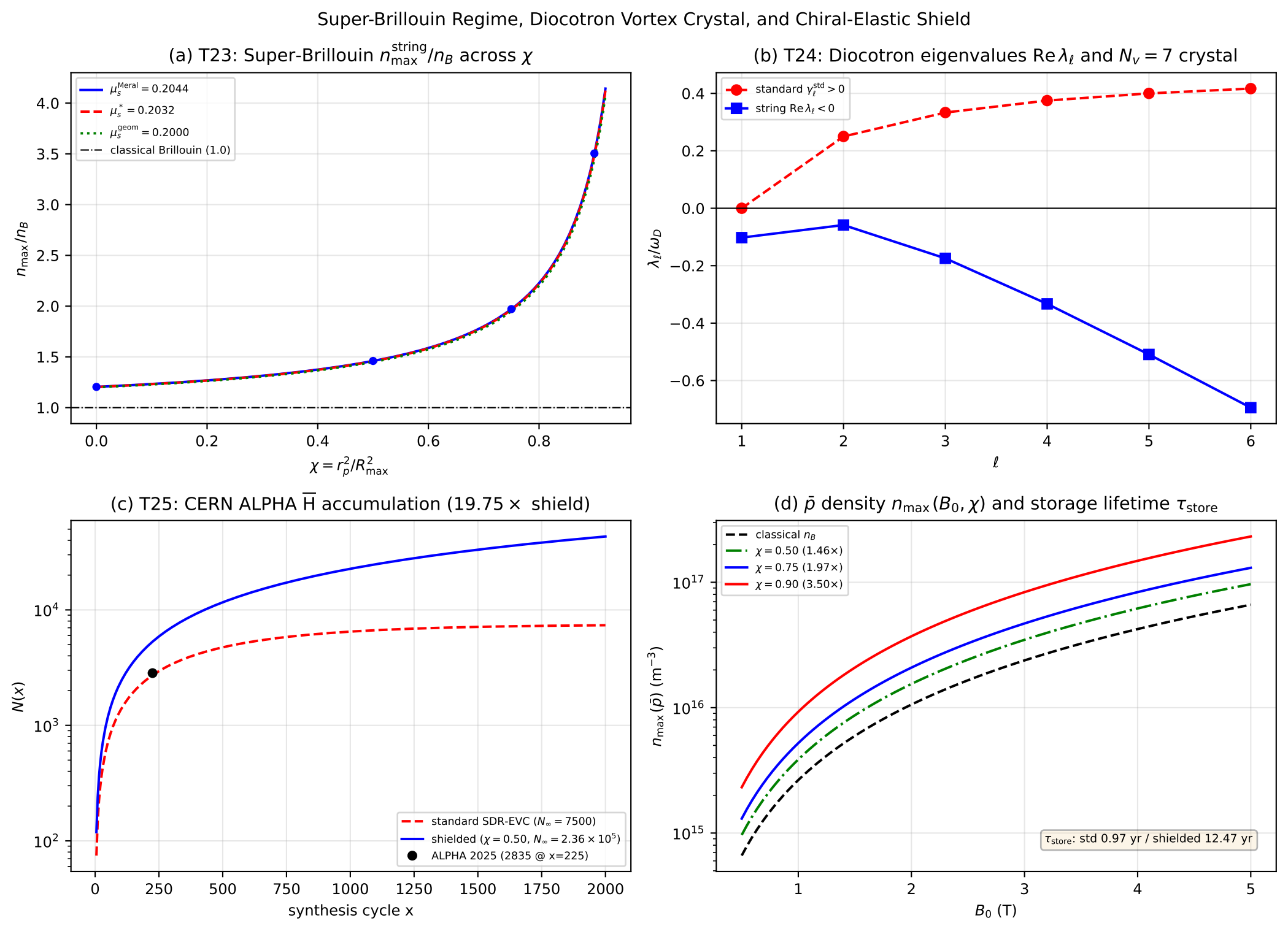
Super-Brillouin Regime, Diocotron Vortex Crystal, and Chiral-Elastic Shield
(a) T23: Super-Brillouin $n_{\max}^{\text{string}}/n_B$ across χ
$\mu_s^{\text{Meral}} = 0.2044$
$\mu_s^{*} = 0.2032$
$\mu_s^{\text{geom}} = 0.2000$
classical Brillouin (1.0)
$n_{\max}/n_B$
$\chi = r_p^2/R_{\max}^2$
(b) T24: Diocotron eigenvalues $\mathrm{Re}\,\lambda_\ell$ and $N_v = 7$ crystal
standard $\gamma_\ell^{\text{std}} > 0$
string $\mathrm{Re}\,\lambda_\ell < 0$
λ_ℓ/ω_D
ℓ
(c) T25: CERN ALPHA $\overline{\text{H}}$ accumulation (19.75 × shield)
standard SDR-EVC ($N_\infty = 7500$)
shielded ($\chi = 0.50$, $N_\infty = 2.36 \times 10^5$)
ALPHA 2025 (2835 @ x=225)
$N(x)$
synthesis cycle x
(d) $\bar{p}$ density $n_{\max}(B_0, \chi)$ and storage lifetime τ_{store}
classical n_B
$\chi = 0.50$ (1.46×)
$\chi = 0.75$ (1.97×)
$\chi = 0.90$ (3.50×)
τ_{store}: std 0.97 yr / shielded 12.47 yr
$n_{\max}(\bar{p})$ (m^{-3})
B_0 (T)